

## SAFETY DATA SHEET

Revision Date 01-Apr-2024

Revision Number 4

### 1. Identification

**Product Name** Copper etchant

**Cat No. :** 44583

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                   |            |
|-----------------------------------|------------|
| Corrosive to metals               | Category 1 |
| Acute oral toxicity               | Category 4 |
| Skin Corrosion/Irritation         | Category 2 |
| Serious Eye Damage/Eye Irritation | Category 1 |
| Skin Sensitization                | Category 1 |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

May be corrosive to metals  
Harmful if swallowed  
Causes skin irritation  
May cause an allergic skin reaction

Causes serious eye damage



### Precautionary Statements

#### Prevention

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Keep only in original container

#### Skin

IF ON SKIN: Wash with plenty of soap and water  
Take off contaminated clothing and wash before reuse  
If skin irritation or rash occurs: Get medical advice/attention

#### Eyes

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER or doctor/physician

#### Ingestion

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell  
Rinse mouth

#### Spills

Absorb spillage to prevent material damage

#### Storage

Store in corrosive resistant polypropylene container with a resistant inliner

#### Disposal

Dispose of contents/container to an approved waste disposal plant

#### Hazards not otherwise classified (HNOC)

None identified

## 3. Composition/Information on Ingredients

| Component          | CAS No    | Weight % |
|--------------------|-----------|----------|
| Water              | 7732-18-5 | 56.90    |
| Iron(III) chloride | 7705-08-0 | 42.00    |
| Hydrochloric acid  | 7647-01-0 | 1.1      |

## 4. First-aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

|                                            |                                                                                                                                                                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                                                                                                                                        |
| <b>Most important symptoms and effects</b> | Causes eye burns. May cause allergic skin reaction. Causes severe eye damage. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                               |

## 5. Fire-fighting measures

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Suitable Extinguishing Media</b>     | Not combustible.         |
| <b>Unsuitable Extinguishing Media</b>   | No information available |
| <b>Flash Point</b>                      | No information available |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

Hydrogen chloride. Iron oxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
3

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
-

## 6. Accidental release measures

|                                  |                                                                                                                                                                  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment as required.                                                                                      |
| <b>Environmental Precautions</b> | Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system. |

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

|                 |                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong bases. Metals. Amines.                                |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component          | ACGIH TLV                | OSHA PEL                                                                                                             | NIOSH                                                          | Mexico OEL (TWA)         |
|--------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------|
| Iron(III) chloride | TWA: 1 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup>                                                                                   | TWA: 1 mg/m <sup>3</sup>                                       | TWA: 1 mg/m <sup>3</sup> |
| Hydrochloric acid  | Ceiling: 2 ppm           | Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup><br>(Vacated) Ceiling: 5 ppm<br>(Vacated) Ceiling: 7 mg/m <sup>3</sup> | IDLH: 50 ppm<br>Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup> | Ceiling: 2 ppm           |

#### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures** Ensure that eyewash stations and safety showers are close to the workstation location.

#### Personal Protective Equipment

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:** Multi-purpose/ABEK. conforming to EN14387. Particulates filter conforming to EN 143.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Liquid                   |
| Appearance                             | No information available |
| Odor                                   | Acidic                   |
| Odor Threshold                         | No information available |
| pH                                     | No information available |
| Melting Point/Range                    | No data available        |
| Boiling Point/Range                    | No information available |
| Flash Point                            | No information available |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | 23 hPa @ 20 °C           |
| Vapor Density                          | No information available |
| Specific Gravity                       | 1.19 g/cm <sup>3</sup>   |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |

## 10. Stability and reactivity

**Reactive Hazard** None known, based on information available

|                                         |                                          |
|-----------------------------------------|------------------------------------------|
| <b>Stability</b>                        | Stable under normal conditions.          |
| <b>Conditions to Avoid</b>              | Incompatible products.                   |
| <b>Incompatible Materials</b>           | Strong bases, Metals, Amines             |
| <b>Hazardous Decomposition Products</b> | Hydrogen chloride, Iron oxides           |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur. |
| <b>Hazardous Reactions</b>              | None under normal processing.            |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

**Oral LD50** Category 4. ATE = 300 - 2000 mg/kg.

**Dermal LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Vapor LC50** Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

#### Component Information

| Component          | LD50 Oral                              | LD50 Dermal             | LC50 Inhalation       |
|--------------------|----------------------------------------|-------------------------|-----------------------|
| Water              | -                                      | -                       | -                     |
| Iron(III) chloride | 450 mg/kg ( Rat )<br>316 mg/kg ( Rat ) | Not listed              | Not listed            |
| Hydrochloric acid  | 238 - 277 mg/kg ( Rat )                | > 5010 mg/kg ( Rabbit ) | 1.68 mg/L ( Rat ) 1 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | No information available                                                                 |
| <b>Sensitization</b>   | May cause sensitization by skin contact                                                  |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component          | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------|-----------|------------|------------|------------|------------|------------|
| Water              | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Iron(III) chloride | 7705-08-0 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Hydrochloric acid  | 7647-01-0 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Contains a substance which is: The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component          | Freshwater Algae | Freshwater Fish                                                                                                              | Microtox   | Water Flea                                                                             |
|--------------------|------------------|------------------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------|
| Iron(III) chloride | Not listed       | LC50: 20.95 - 22.56 mg/L, 96h semi-static (Pimephales promelas)<br>LC50: = 20.26 mg/L, 96h semi-static (Lepomis macrochirus) | Not listed | EC50: = 9.6 mg/L, 48h Static (Daphnia magna)<br>EC50: = 27.9 mg/L, 48h (Daphnia magna) |
| Hydrochloric acid  | -                | 282 mg/L LC50 96 h Gambusia affinis<br>mg/L LC50 48 h Leuciscus idus                                                         | -          | 56mg/L EC50 72h Daphnia                                                                |

**Persistence and Degradability** May persist based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

| Component          | log Pow |
|--------------------|---------|
| Iron(III) chloride | -4      |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN3264  
 Proper Shipping Name Corrosive liquid, acidic, inorganic, n.o.s.  
 Technical Name (FERRIC CHLORIDE, HYDROCHLORIC ACID)  
 Hazard Class 8  
 Packing Group III

### TDG

UN-No UN3264  
 Proper Shipping Name Corrosive liquid, acidic, inorganic, n.o.s.  
 Hazard Class 8  
 Packing Group III

### IATA

UN-No UN3264  
 Proper Shipping Name Corrosive liquid, acidic, inorganic, n.o.s.  
 Hazard Class 8  
 Packing Group III

### IMDG/IMO

UN-No UN3264  
 Proper Shipping Name Corrosive liquid, acidic, inorganic, n.o.s.  
 Hazard Class 8  
 Packing Group III

## 15. Regulatory information

### United States of America Inventory

| Component | CAS No | TSCA | TSCA Inventory notification - | TSCA - EPA Regulatory |
|-----------|--------|------|-------------------------------|-----------------------|
|-----------|--------|------|-------------------------------|-----------------------|

|                    |           |   | Active-Inactive | Flags |
|--------------------|-----------|---|-----------------|-------|
| Water              | 7732-18-5 | X | ACTIVE          | -     |
| Iron(III) chloride | 7705-08-0 | X | ACTIVE          | -     |
| Hydrochloric acid  | 7647-01-0 | X | ACTIVE          | -     |

**Legend:**

TSCA - US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component          | CAS No    | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|--------------------|-----------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Water              | 7732-18-5 | X   | -   | 231-791-2 | X     | X    |      | X    | X     | KE-35400 |
| Iron(III) chloride | 7705-08-0 | X   | -   | 231-729-4 | X     | X    | X    | X    | X     | KE-21134 |
| Hydrochloric acid  | 7647-01-0 | X   | -   | 231-595-7 | X     | X    | X    | X    | X     | KE-20189 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component         | CAS No    | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|-------------------|-----------|----------|-------------------------------|---------------------------------|
| Hydrochloric acid | 7647-01-0 | 1.1      | 1.0 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

| Component          | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|--------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Iron(III) chloride | X                          | 1000 lb                     | -                      | -                         |
| Hydrochloric acid  | X                          | 5000 lb                     | -                      | -                         |

**Clean Air Act**

| Component         | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|-------------------|-----------|-------------------------|-------------------------|
| Hydrochloric acid | X         |                         | -                       |

OSHA - Occupational Safety and Health Administration

Not applicable

| Component         | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|-------------------|----------------------------------|----------------------------|
| Hydrochloric acid | -                                | TQ: 5000 lb                |

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component          | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances RQs | SARA Reportable Quantity (RQ) |
|--------------------|--------------------------|-------------------------------------------|-------------------------------|
| Iron(III) chloride | 1000 lb                  | -                                         | 1000 lb<br>454 kg             |
| Hydrochloric acid  | 5000 lb                  | 5000 lb                                   | 5000 lb<br>2270 kg            |

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component          | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|--------------------|---------------|------------|--------------|----------|--------------|
| Water              | -             | -          | X            | -        | -            |
| Iron(III) chloride | X             | X          | X            | -        | X            |
| Hydrochloric acid  | X             | X          | X            | X        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y  
 DOT Marine Pollutant N  
 DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product contains the following DHS chemicals:

**Legend** - STQs = Screening Threshold Quantities, APA = A placarded amount

| Component         | DHS Chemical Facility Anti-Terrorism Standard                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------|
| Hydrochloric acid | Release STQs - 15000lb (concentration >=37%)<br>Release STQs - 5000lb (anhydrous)<br>Theft STQs - 500lb (anhydrous) |

**Other International Regulations****Mexico - Grade**

No information available

**Authorisation/Restrictions according to EU REACH**

| Component          | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Water              | 7732-18-5 | -                                                                   | -                                                                             | -                                                                                                     |
| Iron(III) chloride | 7705-08-0 | -                                                                   | -                                                                             | -                                                                                                     |
| Hydrochloric acid  | 7647-01-0 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

**REACH links**<https://echa.europa.eu/substances-restricted-under-reach>**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component          | CAS No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Water              | 7732-18-5 | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Iron(III) chloride | 7705-08-0 | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Hydrochloric acid  | 7647-01-0 | Listed   | Not applicable               | Not applicable            | Not applicable                             |



Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

#### Other International Regulations

| Component          | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water              | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Iron(III) chloride | 7705-08-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Hydrochloric acid  | 7647-01-0 | 25 tonne                                                                                  | 250 tonne                                                                                | Not applicable             | Annex I - Y34                      |

### 16. Other information

**Prepared By**

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Revision Date**

01-Apr-2024

**Print Date**

01-Apr-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**